

BLOEMFONTEIN INTL, South AfricaWMO: **684420**Lat: **29.1039S**Lon: **26.2981E**Elev: **1353**StdP: **86.09**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-4.8	-3.4	-15.0	1.2	16.4	-12.7	1.5	13.8	9.7	13.9	8.4	14.0	0.7	210	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.4	34.1	15.4	32.9	15.4	31.8	15.4	19.7	26.3	19.1	25.9	18.5	25.4	4.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	15.3	21.6	17.1	14.4	21.1	16.4	13.8	20.8	62.9	25.7	60.4	25.8	58.3	25.3	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.8	6.9	DB	-7.3	36.2	1.5	1.7	-8.4	37.4	-9.2	38.4	-10.1	39.3	-11.2	40.6	(4)
(5)			WB	-8.2	21.2	1.5	1.2	-9.2	22.0	-10.1	22.7	-11.0	23.4	-12.1	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	22.7	22.0	19.8	15.5	11.7	8.2	7.9	10.7	14.9	18.1	19.9	21.9
	DBStd	5.92	2.43	2.23	2.51	2.73	2.78	2.47	2.60	3.37	3.47	3.20	3.27	2.56
	HDD10.0	190	0	0	0	1	16	63	75	31	3	0	0	0
	HDD18.3	1358	2	3	14	89	207	303	324	236	112	43	21	3
	CDD10.0	2411	392	335	304	167	67	11	9	54	150	252	298	371
	CDD18.3	537	136	104	60	6	0	0	0	1	9	37	69	115
	CDH23.3	7219	1490	1087	756	188	31	1	1	57	386	737	1047	1439
	CDH26.7	2776	656	432	238	27	1	0	0	4	106	259	424	629
Wind	WSAvg	2.7	3.2	2.8	2.5	2.1	1.9	1.9	2.0	2.6	2.9	3.3	3.6	3.5
Precipitation	PrecAvg	549	87	94	75	47	20	13	8	13	19	45	64	65
	PrecMax	1117	228	530	201	145	89	95	49	117	189	216	190	163
	PrecMin	282	9	5	16	2	0	0	0	0	0	0	3	3
	PrecStd	183	59	83	39	35	22	17	12	24	31	44	46	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.9	34.2	33.0	29.8	25.9	22.2	22.9	27.1	31.8	33.2	34.2	36.1
		MCWB	15.5	16.3	15.6	14.4	12.3	9.9	9.2	11.7	13.7	14.0	14.9	15.3
	2%	DB	34.0	32.8	31.0	27.1	24.1	20.8	21.0	24.9	29.6	31.2	32.8	33.9
		MCWB	15.5	16.1	15.9	14.3	11.6	9.4	8.8	10.6	12.8	13.6	14.7	15.4
	5%	DB	32.2	31.2	29.2	25.6	22.7	19.1	19.5	23.1	27.7	29.8	31.1	32.0
		MCWB	16.2	16.4	16.0	13.9	11.2	8.7	8.4	10.0	11.8	13.4	14.5	15.3
	10%	DB	30.6	29.7	27.8	24.0	21.0	17.9	17.9	21.1	25.5	27.9	29.2	30.2
		MCWB	16.5	16.8	16.1	13.7	11.0	8.2	8.0	9.3	11.3	13.2	14.2	15.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	20.6	19.5	17.6	14.6	11.9	11.3	13.5	15.8	17.6	18.5	20.1
		MCDB	27.1	27.5	26.2	23.6	21.3	18.6	18.8	23.2	26.7	25.1	25.8	27.3
	2%	WB	19.7	19.7	18.6	16.4	13.5	10.7	10.1	12.2	14.3	16.3	17.6	19.0
		MCDB	26.0	26.3	25.5	22.7	20.4	18.1	18.5	21.6	25.2	24.6	25.2	26.4
	5%	WB	19.1	19.1	17.9	15.6	12.5	9.7	9.1	11.2	13.3	15.5	16.9	18.1
		MCDB	25.7	25.8	25.0	22.2	19.6	17.2	17.8	20.5	24.0	24.1	24.9	25.9
	10%	WB	18.4	18.4	17.3	14.8	11.6	8.7	8.2	10.1	12.2	14.7	16.2	17.5
		MCDB	25.2	25.0	24.4	21.5	18.8	16.1	16.9	19.1	22.7	23.5	24.2	25.3
Mean Daily Temperature Range	5% DB	MDBR	14.4	14.0	14.8	15.9	17.7	18.6	19.3	19.4	19.3	17.3	16.3	15.6
		MCDBR	17.7	17.5	17.8	18.6	20.5	20.8	21.5	21.7	22.5	20.3	18.9	18.6
		MCWBR	6.0	6.0	6.6	8.0	9.7	11.1	11.4	10.7	10.0	8.1	7.0	6.7
	5% WB	MCDBR	13.0	12.9	14.1	14.6	16.5	17.8	19.2	18.5	18.5	15.8	15.0	14.7
	MCWBR	5.1	5.2	5.9	7.2	8.5	10.0	10.8	9.7	9.0	6.9	6.4	6.1	
Clear-Sky Solar Irradiance	taub	0.345	0.350	0.340	0.304	0.281	0.265	0.264	0.315	0.351	0.368	0.328	0.331	
	taud	2.434	2.422	2.422	2.504	2.546	2.556	2.546	2.384	2.276	2.268	2.432	2.463	
	Ebn at Noon	996	973	950	939	918	913	930	911	922	943	1006	1012	
	Edh at Noon	123	121	114	94	81	76	80	105	129	139	122	120	
All-Sky Solar Radiation	RadAvg	7.47	6.71	5.89	4.78	4.03	3.60	3.94	4.77	5.92	6.85	7.62	7.87	
	RadStd	0.59	0.62	0.33	0.28	0.21	0.18	0.14	0.26	0.26	0.36	0.44	0.41	

Historical Trends

	DBAvg	Heating	
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